NEBIUS × NVIDIA GLOBAL AI HACKATHON · SYSTEM ARCHITECTURE

StutterClear · your real voice, without the repeats

Stutter removal in the speaker's own voice, live on calls and in recordings, with NVIDIA Nemotron checking the meaning of every cut. As deployed on 26 Oct 2026 (commit `ab3eb2b`); every number is measured on 7 speakers the models never trained on (23 min of real stuttered speech).

33% → 77% of stutters removed live (GPU model, Gentle at 50 → Strong at 100)

91% · 0 removed from recordings (Thorough) · real words sliced by a cut

109 s · 4.7 s free GPU waking on demand · to analyse an 18.5 s voice note once awake

5 roles for NVIDIA Nemotron 3 Super on Nebius Token Factory, text only

Who uses it

any modern browser · phones

Visitor's browser — React 19 · Vite

- Tabs: **Calls** (live) · **Recordings** · Dictate · On stage · Practice
- Mic captured by an **AudioWorklet at 16 kHz**, sent as **20 ms float32 frames**
- **Fluency Mirror**: what you said, what was cut, what the listener hears; live captions
- **Sensitivity bar 0–100**, changed while you talk
- A/B: Original · Clean voice · As the call heard it
- Fluency aid and review feedback stay in the browser

Front door — GitHub Pages

- The link judges open: a static page that **health-checks each server** and opens the first that answers
- If none answers: video, results, status, **never an error page**

On your own computer — local mode

- Cleaned voice to headphones or a **virtual mic for Zoom, Meet, Teams**
- Big model on the laptop GPU (RTX 4050, **≈55 ms a window**), 0.54 s behind you
- Can also **stand by as the GPU worker**

Built with people who stutter in mind

- Always **your own voice**; Say it is labelled synthetic
- Word rescue **suggests**; it never speaks for you
- No diagnosis, no cure claims

NVIDIA models & outside services

called over HTTPS; keys live only in Secret Manager

NVIDIA Nemotron 3 Super — Nebius Token Factory

meaning guard · word rescue · context agent · insights · interview

`nemotron-3-super-120b-a12b` · text only, never audio · keeps meant repeats ("that that is true")

NVIDIA Magpie TTS — build.nvidia.com

"Say it": speaks a word you picked, into the call or on stage; always labelled as a synthetic voice

Tavily

Web search for the context agent: names and terms you may need

Hugging Face Hub

WavLM-Large and CrisperWhisper 2.0 at **pinned revisions**; the worker downloads them at boot

GitHub — public repo

The worker runs **exactly the server's commit**. Actions: every PR runs the tests **Karma** selects; main runs the full suite

HTTPS · text only

models

code at the commit

/ws/mic 20 ms audio

/ws: mirror, captions

health check, then opens

App server · Google Cloud Run, Mumbai

FastAPI + uvicorn, one process · 2 vCPU, 4 GiB, CPU PyTorch · max 1 instance, session affinity

Live pipeline — /ws/mic → /ws

20 ms frames → detector → splicer → listener

- **Detector**: rule-based on the CPU (DTW repeat match, steady-sound prolongations), or **Best (GPU)**: teacher windows sent to the worker, a rule-based stand-in running alongside
- **Splicer**: delay 0.25–1.14 s; never plays audio not yet judged; skips cut spans with 8 ms fades; the room's own tone, never digital silence; lag rebuilt in natural pauses
- **Sensitivity** scales every detector's thresholds; cut limits grow with it
- Captions: NVIDIA **FastConformer** streaming (sherpa-onnx, int8) on the CPU

Recording engines — Natural · Thorough · Live

teacher probabilities · verbatim words → word-aligned cuts → Nemotron guard

- Cuts land **only between words** (CrisperWhisper timings): no real word is sliced
- Nemotron reviews every proposed cut and keeps the repeats that were meant
- Thorough also cuts um, uh and repeated phrases, and shrinks long pauses to under 0.5 s
- Rendered with crossfades and room tone; **Live** on a recording: the live model's cuts with the time closed up

Features on top

- **Voice notes** ≤ 3 min (1 h in memory, never on disk) · **Dictate** (cleaned text) · **Clean my video** (ffmpeg cuts picture and sound together)
- **Demo clips**: 3 real UCLASS recordings, private in the image and hash-checked, with cached model output and the GPU model's recorded live decisions, replayed while the GPU sleeps
- Word rescue · Context card · Insights · Interview practice · Say it

GPU broker — wakes on demand

- Wakes a GPU **only on an action**: voice note, Dictate, video, Best (GPU) live, Wake button
- Each wake gets its own **signed token** (HMAC); the worker must present it
- Queue with fair shares; **live windows never wait behind a recording**
- Budget: ≤ 20 of Kaggle's ~30 GPU hours a week, ≤ 5 a day; sleeps 10 min after its last job

Guards · same-origin checks · per-route body limits · download budgets · per-visitor limits · one live slot (5 min a session, 20 min an hour each) · nothing stored on disk · secrets from Secret Manager

Kaggle API: push notebook

wss /ws/gpu live + jobs

GPU worker · Kaggle, 2× T4

or the team laptop

A wake, measured on 26 Oct

6 s	26 s	77 s	109 s
notebook pushed	boot, code at the exact commit, locked packages	models fetched, loaded, warmed	ready

Connects **out** to the app: nothing is opened where it runs. Leaves by itself when idle.

GPU 0 · live teacher (v1)

- **WavLM-Large + BiGRU** head, 5 classes per 20 ms frame
- Sliding windows: 4 s of context + **200 ms lookahead** (Gentle) or 500 ms (Strong); a decision every 160 ms
- Round trip from Mumbai: p90 ≈ 0.3 s, so Best (GPU) runs 0.84 s behind you

GPU 1 · recordings

- Teacher v2 on 30 s chunks, fp16
- **CrisperWhisper 2.0 large**: verbatim words with timings, stutters included; encoder run without attention maps (4.0 GB peak, not 6.8)
- **18.5 s of audio analysed in 4.7 s**

Only GPU work runs here. Cut logic, the meaning guard, captions and rendering stay on the app server. Weights are mounted read-only from private Kaggle notebooks and never published; nothing is kept on the machine.

A live call, frame by frame

1. **Mic** — 20 ms frames at 16 kHz, from the browser or the laptop's device
2. **Delay buffer** — your voice waits 0.25 s (rule-based) to 0.84 s (GPU model, Gentle)
3. **Detector** — judges each 20 ms frame once it has heard 200 ms after it (500 ms in Strong)
4. **Splicer** — plays only judged audio, skips each stutter span; while one is still going it holds the room's tone
5. **Listener** — hears your own voice. A removed stutter becomes a short pause (a call can't play what wasn't said); recordings close it up

Sensitivity — 0: clear stutters only · 50: as measured · 100: cuts aggressively

GPU model, Gentle: removes **12% → 33% → 60%** of stutters at 0 / 50 / 100 (0.6% → 3.1% → 10% of words lost); Strong reaches **77%**. Above 50 a cut may run longer before it counts as a detector error.

Models & training

- **Teacher** (v1 live, v2 recordings): WavLM-Large + BiGRU head → per 20 ms frame: sound repetition, word repetition, prolongation, interjection, and block (reported, never cut)
- **Trained on Kaggle GPUs**: SEP-28k, FluencyBank, LibriSpeech with synthetic stutters spliced in, and **silver labels from CrisperWhisper** verbatim transcripts of real stuttered speech
- **Causal student** (CNN + GRU, ONNX int8) for laptops without a GPU; **rule-based** detector as the fallback that always runs
- **SEP-28k-E** (speaker-exclusive): macro F1 **0.618**; AUC 0.98 for word repetitions, 0.96 for fillers

Measured on 7 held-out speakers

	STUTTERS REMOVED	WORDS LOST
Live, rule-based (no GPU)	7.6%	1.25%
Live, GPU, Gentle	**33.1%**	3.10%
Live, GPU, Strong (fillers too)	**65.2%**	5.68%
… at sensitivity 100: Gentle / Strong	59.5% / **77.3%**	9.98% / 12.96%
Recordings: Natural / Thorough	57.2% / **90.9%**	0 words sliced

Scored like `scripts/eval_live.py`: each recording streamed 10 ms at a time through the real pipeline, against verbatim labels. Live delay 0.25–1.14 s.

Trust, privacy & delivery

- **Stores nothing**: voice notes live 1 h in memory; a live session's data goes only to its own tab
- Real recordings (UCLASS, research licence, credited) and model checkpoints are **never public**: private image, private notebooks
- **Supply chain**: hash-locked packages, pinned base image, one image per commit, deployed by digest; the worker runs only that commit
- **Delivery**: Cloud Build → Artifact Registry → Cloud Run; GitHub Actions + Karma on every PR

live audio & updates · GPU work · NVIDIA model calls · control, code & models

github.com/Pradyuman-aviator/stutterClear · try it: pradyuman-aviator.github.io/stutterClear · sources: README.md, docs/DEPLOY.md, Cloud Run logs